Velvadapu Naga Venkata Ganeswar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.60
Sri Chaitanya Junior College	Class XII (Board of Intermediate Education, Andhra Pradesh)	2023	97.50%
Dr. K.K.R's Gowtham High School	Class X (Borad of Secondary Education, Andra Pradesh)	2021	99.30%

EXPERIENCE

Software Development Intern - Fletch

- Developed reusable, configurable **React** components for a no-code Widget Builder platform, integrating them with REST APIs and a **configuration-driven dynamic rendering architecture.**
- Implemented **custom domain mapping** using **Python** and AWS APIs, automating **TLS certificate provisioning** through **AWS Certificate Manager** and configuring **CloudFront distributions** for HTTPS delivery.
- Migrated frontend modules from **JavaScript to TypeScript** by defining **type-safe API contracts**, component props, state, and widget configuration interfaces across the React application.
- Maintained **CI/CD workflows** for frontend deployments, automating production builds and deployment to **Amazon S3** with **CloudFront** cache invalidation.

Web Developer | MSME, Civil Departments

- Integrated and deployed **Payload CMS** to streamline **content management** across departmental and research laboratory websites.
- Containerized web applications using **Docker** and configured **NGINX** reverse proxy deployments on private campus infrastructure.
- Implemented **Jenkins CI/CD** pipelines to automate application builds, deployments, and release workflows.

PROJECTS

Reliable UDP File Transfer Protocol | C++

- Developed a reliable file transfer protocol over **UDP** in C++, implementing packet segmentation, custom protocol headers, **transaction IDs**, **sequence numbers**, and **acknowledgments (ACKs).**
- Implemented **Selective Repeat ARQ** with **sliding-window flow control**, **per-packet timeout detection**, duplicate packet handling, and **selective retransmission** for reliable data delivery.
- Designed **out-of-order packet buffering** and **file reassembly** mechanisms to ensure correct file reconstruction under **packet loss**, duplication, and **reordering**.
- Benchmarked the UDP protocol against **TCP** under varying file sizes and network conditions, analyzing **throughput**, **transfer time**, **packet retransmissions**, and **protocol overhead**.

Key Management Software | Go, React, Tailwind CSS, PostgreSQL

- Developed a full-stack key management system using **Go, React, Tailwind CSS, and PostgreSQL** for managing 500+ rooms and 1,000+ keys across 3 residential blocks.
- Integrated **OAuth 2.0** authentication and implemented **Role-Based Access Control (RBAC)** with 5 user roles for role-specific access to key management operations.
- Designed a **normalized relational database schema** in PostgreSQL for rooms, keys, users, roles, and key transactions, defining relationships and constraints for consistent data management.

SKILLS

Languages: Go, Python, Dart, C/C++, SQL (PostgreSQL), JavaScript, HTML/CSS

Frameworks: Go/Gin, Flutter, React, Express.js, FastAPI, Next.js, Node.js

Tools: Git, Docker, AWS, PostgreSQL, Jenkins

POSITIONS OF RESPONSIBILITY

Overall Head & Core Member | Lambda, IIT Hyderabad

- Led development of a campus-wide key management system tracking **1,000+ physical keys** across **500+ rooms** and **3 buildings**, streamlining key issuance, returns, and administrative access management.
- Led development of the IITH Dashboard marketplace, facilitating **150+ listings** and streamlining campus-wide buying and selling.
- Organized **10+ technical sessions** and workshops, engaging **200+ students** during the tenure.
- Led merchandise operations, facilitating the sale of **50+ units** across Lambda initiatives and events.

Head of Web Development & Web Manager | E-Cell, IIT Hyderabad

- Automated E-MERGE registration workflows using **Google Apps Script**, processing **200+ registrations** and automatically recording submission data in Google Sheets.
- Built a Startup Studio platform connecting **25+ startups** with **10+ mentors** across **3+ domains**, streamlining mentor discovery and engagement.
- Developed an investment simulation platform used by **100+ participants** to simulate investments across **20+ startups** during E-Cell flagship events.

ACHIEVEMENTS

MILAN Hackathon - 3rd Place

- Secured 3rd place in MILAN, a hackathon conducted by Lambda IIT Hyderabad, for building a Community App for IIT Hyderabad